I. Surplus interconnection service is an important tool to bring new electricity supply online to meet rapidly growing demand. (Roselle)

Electricity demand in the United States is rapidly rising for the first time in decades. ICF, a leading energy consulting and analysis firm, estimates that "U.S. electricity demand is . . . expected to grow 25% by 2030 and 78% by 2050 from 2023 levels."[1] [sentence about data center growth in particular]

At the same time, new electricity supply is not connecting to the grid quickly enough to meet this new demand. Before new power plants and batteries may connect to the grid, they must obtain an Interconnection Agreement from the grid operator, after completing studies to ensure that the right grid infrastructure is in place to reliably accommodate the new electricity. Where the grid isn't adequate, new infrastructure upgrades must first be built before the new resource can interconnect. This process is currently proceeding far too slowly, with new electricity supply resources averaging over 40 months in PJM and MISO from the time they submit interconnection requests to the time they receive interconnection agreements.[2] It then generally takes more than 2 years from the time a resource obtains an interconnection agreement to the time when it reaches commercial operation,[3] or even longer where grid upgrades are required, meaning that the process as a whole of connecting a new power plant averages more than 6 years.

With new demand rising and supply snarled in this 'interconnection queue' process, customers suffer. Prices are skyrocketing, and without new supply, grid reliability could suffer or customers may not obtain adequate service. MISO's most recent planning reserve auction yielded prices of $666.50/MW-day for the summer across all regions, more than 22 times higher than prices the previous year.[4] Similar, the most recent PJM capacity auction returned its maximum prices of $329.17/MW-day.[5] And because it projects that capacity supply will not be adequate to meet demand in the future, PJM recently proposed a process to mandate curtailment for new large load customers. Even the potential that new supply will not come online fast enough to meet new demand threatens Indiana's

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[1] Lalit Batra et al., *Rising current: America's growing electricity demand*, at 3 (June 2025), available at https://www.icf.com/-/media/files/icf/reports/2025/energy-demand-report-icf-2025_report.pdf?rev=c87f111ab97f481a8fe3d3148a372f7f

[2] [LBNL Queued Up data file, at sheet 28, available at https://emp.lbl.gov/queues.]

[3] *Id.* at sheets 32-33.

[4] *Compare* MISO 2025 PRA Results, available at https://www.misoenergy.org/meet-miso/media-center/2025---news-releases/misos-planning-resource-auction-indicates-sufficient-resources/, with MISO 2024 PRA results, available at https://cdn.misoenergy.org/2024%20PRA%20Results%20Posting%2020240425632665.pdf .

[5] *See* https://www.pjm.com/-/media/DotCom/about-pjm/newsroom/2025-releases/20250722-pjm-auction-procures-134311-mw-of-generation-resources-supply-responds-to-price-signal.pdf

economic growth and compromises national security by weakening the nation's position in the AI race.[6]

Surplus interconnection is an important tool to address these challenges because it offers a pathway to connecting new electricity supply resources to the system more quickly. With surplus interconnection, a new power plant or battery shares grid access with an existing resource. As explained in a recent GridLab report, "[m]any existing power plants do not fully utilize the infrastructure that was built to ensure that their electricity is deliverable to customers, leaving 'surplus' capacity unused on the grid."[7] By promising to inject electricity only when the existing plant isn't using its full allotted output, a new power plant or battery can obtain an interconnection agreement without having to proceed through the standard interconnection queue, cutting the process from many years to a matter of months,[8] while minimizing the need for new grid infrastructure to accommodate the new resource.[9]

Where grid operators adopt rules to facilitate surplus interconnection service and allow these resources to contribute to the grid's capacity needs, surplus interconnection can be an important tool to enhance system reliability and affordability. Recent grid operator reforms have enabled surplus interconnection, especially in the MISO region. But a number of barriers limit its use and, critically, states and the utilities that they regulate must play an integral role in unlocking surplus interconnection to deploy new energy supply.

This report begins with further context on Indiana's energy supply challenge. It discusses the PJM and MISO rules for surplus interconnection, as well further steps that PJM must take to fully enable the practice. We then detail steps that Indiana utilities, regulators, and legislators can take to ensure the rapid development of surplus interconnection resources.

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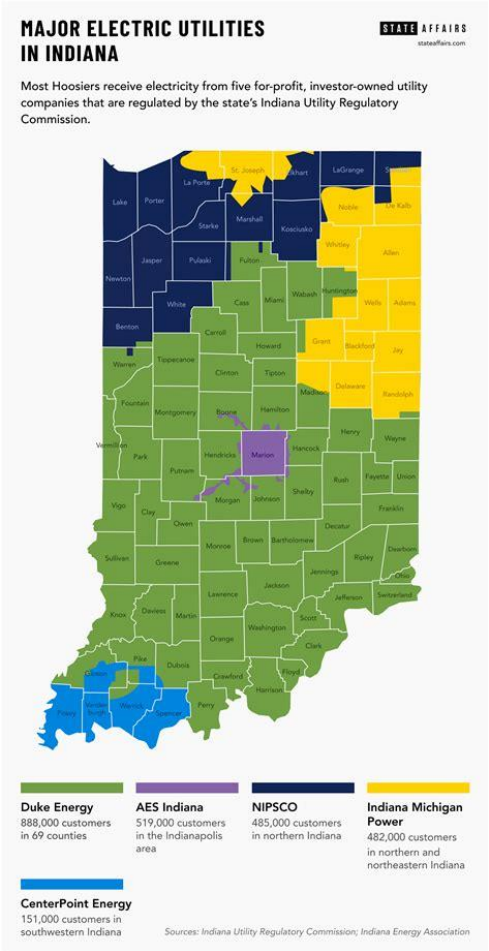
[6] [Add cite]
[7] Miles Farmer & Abe Silverman, *Unlocking the Power of Surplus Interconnection: Barriers, Opportunities, and Strategic Solutions*, at 3 (Feb. 21, 2025) (Hereinafter "Surplus Interconnection: Barriers and Solutions").
[8] *Id.*
[9] *Id.* at 7-8.

II. Context – Indiana's energy supply challenge (Abe)

Indiana has been a leader in creative and business-friendly approaches to managing the electric sector. For example, Indiana's recent Senate Bill 422 embedded the use of Alternative Transmission Technologies into utility planning and serves as a national example of pragmatic, forward-looking leadership. [Possibly include SB 422 as a call-out box?]



Additionally, resource adequacy in Indiana is undergoing a sea-change, with the passage of House Enrolled Act 1007 in 2023. The legislation established "Five Pillars" to guide future IURC review of utility procurements plans.[10] The five pillars are Reliability, Affordability, Resiliency, Stability and Environmental Sustainability. Going forward, the IURC will have to "take each of these attributes into account" when evaluating an IRP, issuance of a CPCN, or upon construction, purchase or lease of a new power plant. Further, HEA 1007 mandates that, post-2023, each IOUs can procure no more than 15% of its UCAP needs from the capacity market, meaning that it must self-supply the remaining 85%. The new statute has created a wave of new IRP proposals that comply with the new legislation.

Indiana is a traditionally regulated, vertically-integrated state with five IOUs:

- Duke Energy Indiana (DEA)
- AES Indiana,
- NIPSCO,
- Center Point, and
- Indiana Michigan Power (AEP).

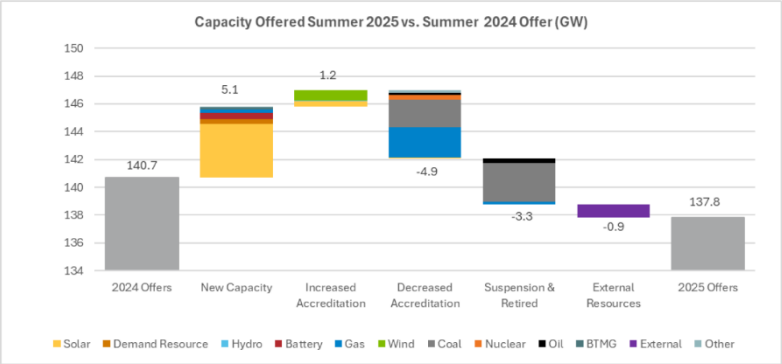
AEP operates in the PJM footprint, while the remaining IOUs are part of MISO. Indiana also has close to 100 municipal and cooperative utilities that span both PJM and MISO, largely

[10] https://www.in.gov/iurc/files/GAO-2023-04_ORDER_06-28-2023.pdf

supplied by the Indiana Municipal Power Agency (IMPA), Hoosier Energy and Wabash Valley Power Alliance (WVPA).

Capacity reserves across the MISO-North and Central regions have continued to tighten from approximately 6.5 GW in 2023, to 4.6 GW in 2024 and down to 2.6 GW in 2025. Capacity prices in Indiana now exceeding $200/MW-Day on an annualized basis and reaching $666.50/MW-day for the Summer 2025 period. Summer peak demand is expected to grow by around 10% between now and 2030 [confirm and add cite]. As MISO notes, "[p]ressure persists with reduced capacity surplus across the region and is reflected through improved price signals in this year's auction" and concluding that the "results reinforce the need to increase capacity, as demand is expected to grow with new large load additions".[11]

New capacity additions did not keep pace with decreased accreditation, suspension retirements and external resources



The role of solar energy in meeting MISO's resource adequacy needs is also changing. By July 2025, MISO expects to have approximately 15 GW of solar operating, up from 6 GW as recently as Summer 2024. MISO officials have highlighted this change as being favorable for new energy storage which can help manage the morning and evening peaks.[12] The avalanche of new solar is also coinciding with regulatory changes that are driving tighter supply-demand conditions. The 2025/2026 auction was the first to be held under MISO's new Reliability-Based Demand Curve. Further, MISO is in the process of updating its capacity accreditation rules. While thermal resources are already operating under the Seasonal Accredited Capacity rules, the new Direct Loss of Load accreditation for non-thermal resources is scheduled to take effect in 2028/2029, and the lower accreditation values are not currently reflected in MISO's supply-demand balance. Thus, while MISO currently assigns solar a relatively high accreditation value (50% for Spring, Summer and Fall, and 5% for Winter), that value is expected to fall significantly with the new DLOL process, further tightening the supply-demand balance.

Resource Adequacy Situation In Indiana

[11] MISO "2025 PRA Result & Summer Readiness", Update to the Indiana Regulatory Utility Commission, May 20, 2025, at p. 9.
[12] MISO Presentation to IRUC, May 20, 2025. 7

Indiana utilities are widely regarded as meeting their resource adequacy needs primarily from owned and contracted resources, with approximately 90% of the state's load hedged. Several utilities report relying on the MISO PRA for a portion of their capacity needs, but all identify being in compliance with HEA 1007's limit of purchasing no more than 15% of the utility's resource adequacy needs from the PRA.

[See appendix for more detailed information on Indiana utilities' RA positions.]

Upcoming Indiana RFP Cycle:

Recent IRPs (filed) & what's next

Utility	Latest IRP filing	What it means	Next up (expected cadence)
Duke Energy Indiana	**2024 IRP** (filed Nov 1, 2024)	Portfolio reflects MISO seasonal accreditation and accelerating load/economic development; continues Edwardsport gas-only option analysis.	IRPs every ~3 yrs ⇒ **~2027** cycle. (Indiana.gov)
AES Indiana	2022 IRP (most recent completed); **2025 IRP process resumed**	Updating plan post-HEA 1007; stakeholder meetings active in 2025.	Target **2025** filing. (AES Indiana)
NIPSCO	**2024 IRP** (filed Dec 9, 2024)	Advances coal retirement path; adds flexible gas/renewables/storage while addressing large-load growth.	**~2027** (statutory 3-yr), though NIPSCO often iterates faster. (Indiana.gov)
CenterPoint IN South	**2023 IRP** on record; **2025 IRP underway** (public advisory Mar 2025)	Reevaluating resource mix & reliability for SW Indiana.	**2025 filing** in progress. (Indiana.gov, CenterPoint Energy, City-County Observer)

Utility	Latest IRP filing	What it means	Next up (expected cadence)
Indiana Michigan Power	**2024 Indiana IRP** (filed Mar 28, 2025)	Plans to meet PJM targets post-Rockport with Cook nuclear + renewables/gas; includes large-load scenarios.	**~2028**. (Indiana Michigan Power)

[insert table of Surplus projects tied out to IRPs]

III. Surplus interconnection processes in MISO and PJM

A. MISO (Abe)

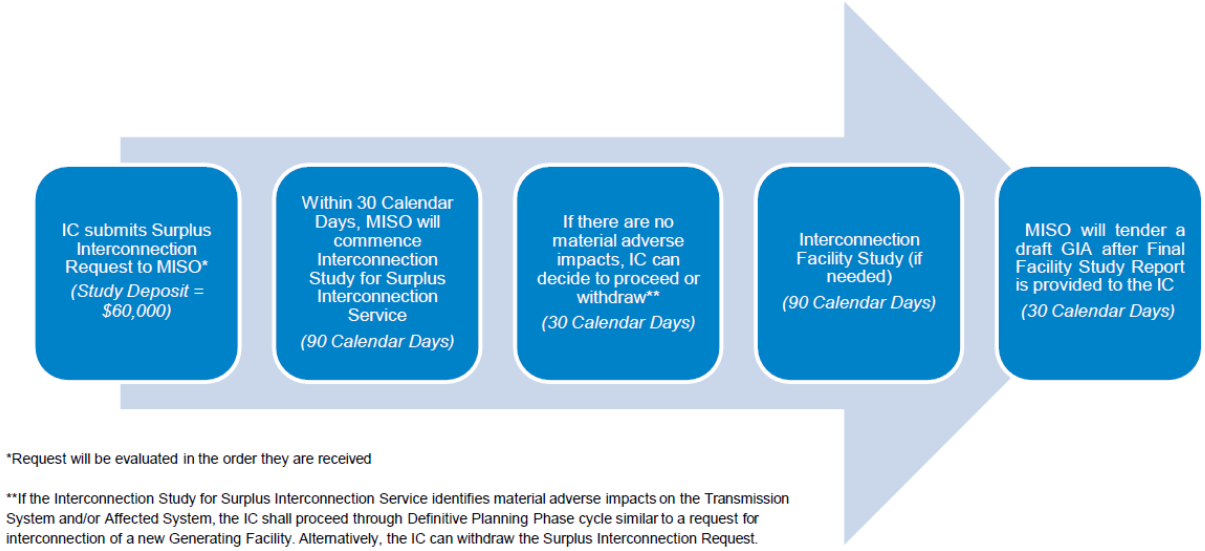
There are currently nine SIS projects located in Indiana, all of them additions of energy storage to existing solar facilities. Projects range in size between 30 MW and 200 MW, with three projects already completed, one withdrawn, and the rest undergoing study. [All] of the projects are part of an IRP process. The majority are servingNIPSCO.

Project Number	Status Reason	Original Fuel Type	New Fuel Type	SIS (MW)	Queue_ID	IRP Linked?	Interconnection Customer (Surplus)	Interconnecting Utility (TO)
S1003	Submitted	Solar	Battery	55	J913	Yes (NIPSCO IRP)		NIPSCO
S1004	Complete	Solar	Storage	30	J903	Unknown		Duke Energy Indiana
S1008	Withdrawn	Solar	Storage	250	J1419	Unknown		Duke Energy Indiana
S1026	Submitted	Solar	Storage	50	J1369	Unknown		CenterPoint Energy (SIGECO)
S1027	Complete	Solar	Storage	50	J1386	Yes (NIPSCO IRP)		NIPSCO
S1028	Complete	Solar	Storage	60	J1067	Yes (NIPSCO IRP)		NIPSCO
S1043	Complete	Solar	Storage	50	J1369	Unknown		CenterPoint Energy (SIGECO)
S1044	Submitted	Solar	Storage	50	J1386	Yes (NIPSCO IRP)		NIPSCO
S1075	Submitted	Solar	Storage	100	J1336	Yes (NIPSCO IRP)		NIPSCO
S1102	Confirmed	Solar	Storage	200	J992	Unknown		Duke Energy Indiana

In MISO, the SIS process is governed by Generator Interconnection Procedures (Attachment X) and lets a new resource use unused interconnection capability at an existing plant's POI, so long as combined injections never exceed the host unit's existing interconnection service.

MISO includes a separate queue for SIS requests, which can be found here: https://cdn.misoenergy.org/Surplus%20Interconnection%20Service%20Requests458123.

xlsx. Study deposit & timeline: MISO's training/implementation deck shows SIS can be requested any time with a $60,000 study deposit. MISO begins the surplus study within 30 days, targets 90 days for the interconnection study, optional 90-day facility study if needed, then tenders a draft GIA. (Att. X 3.3.1.1–3.3.1.3; BPM-015 §6.7.3).



If the host retires: MISO's allows SIS to continue for up to one year after the host unit permanently ceases operations (if requested/ studied accordingly). (Att. X 3.3.1.1–3.3.1.3; BPM-015 §6.7.3).

Two companion agreements are required for each SIS request before the interconnection agreement negotiations conclude:

1. **Monitoring & Consent Agreement (MCA)** (Appendix 11) and
2. **Energy Displacement Agreement (EDA)** (Appendix 12). (Att. X App. 1; App. 11; App. 12).

Ongoing Discussions Regarding Surplus: MISO has an ongoing stakeholder process to consider modifications and enhancements to its SIS process to further streamline its study process,[13] and anticipates filing these modifications in Q1 2026. Discussions surrounding these enhancements demonstrate that while SIS has been a positive tool in MISO, its implementation has encountered certain challenges:

[13] MISO's proposed enhancements include increased study deposits, clarification that MISO will pause any SIS studies if it receives a request to modify the underlying generator (i.e., "Material Modification" requests), a requirement SIS requests be submitted one year prior to the Commercial Operation Date, and allowance for SIS to continue for 1 year after retirement of the existing unit whether that retirement is planned or unplanned. *See* Interconnection Process Working Group (IPWG) materials, July 22, 2025, available at https://www.misoenergy.org/engage/MISO-Dashboard/surplus-interconnection-attachment-x-and-bpm-015-updates/.

- Invenergy has noted that "Existing surplus interconnection studies have been taking over the 270 days depicted in the slide. As of August 12, 2025, using the MISO public posting available in their website, the average number of days for surplus interconnections without study results is 590 days since their queue submittal."

- Earthrise flagged the fact that the replacement study process often takes three years after the host facility stops operating. MISO (and FERC)'s rules, however, provide only a single year for the SIS resource to continue operating post-retirement of the host resource, which creates a potential mismatch in timing that could result in the SIS resource being forced offline.
- NextEra Energy Resources likewise commented that "We encourage extending the one-year limited service concept to situations where "host" LGIAs expire or default for any reason, not just retirements. If the existing generator terminates or defaults on the non-surplus LGIA, the surplus project should be allowed to utilize the generator interconnection process to obtain its own standalone GIA that replaces the terminated or defaulted LGIA without requiring the need to go back into the queue. During this time, the surplus project should be allowed to continue to operate."

If the generator project defaults on its LGIA, the surplus project should still be able to operate in order to offer as much capacity as possible to the MISO system and maintain optimal reliability.

B. PJM (Roselle)

PJM does not have a history of successful development of projects using surplus interconnection service, but this lack of development activity could be poised to change in the near future. The surplus interconnection study process that PJM implemented in the wake of Order No. 845 disqualified all requests so long as they had any impact on the transmission system, effectively disqualifying all commercial use cases of surplus interconnection service.[15] But PJM recently amended its surplus interconnection study rules, providing for surplus interconnection requests to be approved where they do not trigger network upgrades.[16] PJM's new approach, which was enshrined in its business

[15] Surplus Interconnection: Barriers and Solutions, at 19.
[16] *See PJM Interconnection, L.L.C.*, 190 FERC 61,083 (2025) (approving PJM surplus interconnection proposal).

practices manual in Q2 2025,[17] also permits surplus interconnection service even where "Additional Interconnection Facilities . . . are necessary to accommodate the request."[18] These permitted facilities are project-specific infrastructure upgrades that "can be built without adversely impacting the standard interconnection queue."[19] This approach is similar to best practices that have been used successfully in SPP, and potentially allows developers with more flexibility than that permitted under the MISO rules.[20]

Nevertheless, PJM's tariff modifications appear not to have led yet to significant development. PJM does not currently publicly post a surplus interconnection queue like MISO and SPP, but conversations with market participants suggest that, despite FERC's approval of PJM's recent modifications, the region has not yet seen a material uptick in surplus interconnection development. One possible explanation for this is that PJM does not currently have an effective mechanism for transferring interconnection rights from a retiring generator.[21] Where the standard interconnection queue process is delayed and uncertain, such a mechanism is necessary to effectively develop surplus interconnection projects because surplus interconnection rights are tied to the existing facility and expire 1 year after its retirement.[22] PJM recently proposed modifications to its tariff to provide for a separate, expedited interconnection pathway for Capacity Interconnection Rights Transfer.[23] FERC rejected PJM's proposal, leaving the region without a workable pathway for interconnection rights transfer to surplus interconnection resources upon the retirement of the existing resource to which the surplus service is tied.[24] But FERC's order rejecting the proposal clearly identified specific, fixable deficiencies in PJM's proposal that rendered it "unjust and unreasonable" in the Commission's eyes, while otherwise finding that "PJM's proposal would promote the efficient use of existing infrastructure."[25] We

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[17] For stakeholder materials outlining the changes to PJM's business practices manual to implement its revisions to the surplus interconnection study process, see PJM's Markets and Reliability Committee meeting materials for March 19, 2025, available at https://www.pjm.com/committees-and-groups/committees/mrc .

[18] PJM Manual 14H: New Service Requests Cycle Process, at 141 (July 23, 2025).

[19] Surplus Interconnection: Barriers and Solutions, at 21.

[20] *Id.* at 21-22.

[21] While PJM's existing tariff has a Capacity Interconnection Rights Transfer process, flaws in its existing process prevent this pathway from being a workable mechanism for providing a permanent interconnection pathway for surplus interconnection resources. Most significantly, PJM's current process requires replacement resources to "proceed through the new generator interconnection queue, which has historically taken more than five years to complete." FERC Docket No. ER25-1128, Comments in Support of Elevate Renewables F7, LLC, and East Kentucky Power Cooperative, at 6 (Feb. 21, 2025).

[22] Surplus Interconnection: Barriers and Solutions, at 30-33.

[23] *See* FERC Docket No. ER25-1128, Proposed Tariff Amendments for Replacement Generation Interconnection Service (Jan. 31, 2025).

[24] *PJM Interconnection, L.L.C.*, 192 FERC ¶ 61,137, at PP 37-44 (2025).

[25] *Id.* at

anticipate that PJM will propose new revisions that address these issues in the near term, providing a pathway for surplus interconnection projects to convert to permanent interconnection service upon a generator's retirement, thereby unlocking the surplus interconnection development opportunity in PJM.

PJM's market capacity market accreditation, and related market participation and performance rules are also relevant for surplus interconnection resource development, particularly where capacity revenue is important to a project's commercial outlook. PJM's capacity accreditation rules provide a workable pathway for capacity market participation where the existing and new resources are both inverter-based (e.g. solar plus storage). Such resources can participate together with one another in the capacity market. When doing so, they are classified as Combination Resources, and are eligible for ELCC accreditation according to their combination class, or in the case of less typical combinations they may received unit-specific accreditation.[26] In addition, FERC recently approved new market participation rules for storage plus thermal hybrids, which will allow a pathway to unit-specific accreditation for these resources.[27]

However, PJM has not yet amended its rules to allow for other combinations of inverter and non-inverter based resources (e.g. gas plus solar). While this imposes a barrier to solar or wind surplus interconnection resources receiving proper capacity credit when they share interconnection capacity with thermal resources, PJM has indicated a willingness to execute appropriate revisions to allow for such hybrid market participation in the future. In amending its tariff to allow for storage-plus-gas hybrids, PJM explained that it remains open to providing for such modifications, should the market demonstrate that this use case is viable.[28]

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[26] The PJM manual provides for various Hybrid Resource Classes. *See* PJM Manual 21B: PJM Rules and Procedures for Determination of Generating Capacity § 2.3 (July 23, 2025). It also states that "[f]or any resource in an ELCC Class for which no Class rating, the Accredited UCAP shall be based on a resource-specific effective load carrying capability analysis based on the resource's unique parameters." *Id.* § 4.3.

[27] *See PJM Interconnection, L.L.C.*, 190 FERC ¶ 61,195 (2025) (approving PJM's tariff amendments to allow for storage-plus-thermal hybrid market participation). We note that PJM's capacity market manual appears not to have been updated yet to reflect the new tariff, as it continues to state that "[a] mixed-technology resource with a non-inverter generation component and an intermittent component is eligible to participate as Co-Located Resources only." PJM Manual 21B: PJM Rules and Procedures for Determination of Generating Capacity § 2.6 (July 23, 2025). However, we expect that changes to facilitate such participation will be forthcoming shortly.

[28] PJM stated the following in filing its proposal for to allow for hybrid market participation of storage plus non-inverter resources:

> Without existing resources or demonstrations of planned resources with other mixed technologies (e.g., gas plus solar), the proposal in PJM's Distributed Resources Subcommittee could not contemplate appropriate market rules for Hybrid Resources comprised of both inverter and non-

IV. Steps and recommendations for Indiana's utilities to evaluate surplus interconnection opportunities (Abe)

[at the end of this section include utility recommended action, IURC action, and legislative action to implement these steps]

V. Indiana's surplus interconnection opportunity

A. Scale of Indiana's surplus opportunity (essentially a targeted application of the existing GridLab dashboard) (GridLab/ Berkeley)

B. Top surplus interconnection opportunities in Indiana (GridLab / Berkeley analysis based on a screening tool they develop with input from Roselle / Abe)

Researchers at U.C. Berkeley, working in conjunction with GridLab, developed a screening tool to identify the best potential opportunities for surplus interconnection.

Ranking criteria for surplus interconnection development at thermal power plant sites:

1. Economic Arbitrage Opportunity (25% weight): The differential between the plant's variable operating cost and the levelized cost of renewable energy, representing the economic value creation potential from replacing thermal generation with adjacent renewable resources.

2. Renewable Resource Potential (30% weight): The absolute megawatt capacity of combined solar and wind resources available within a 6-mile radius of the plant, indicating the scale of renewable development opportunity.

3. Underutilization Factor (15% weight): The inverse of the plant's capacity factor, where lower utilization rates indicate greater suitability for renewable replacement as these assets are already operating below optimal levels.

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inverter based generation components. PJM will continue to assess the need for further expanding the Hybrid Resources market participation model, as appropriate, to permit the participation of a Hybrid Resource comprised of a non-inverter component and another generation component (e.g., gas plus solar) at a later date as use cases demonstrating how such facilities will operate become more readily apparent. As of now, expanding the market participation model to include such Hybrid Resources would require additional considerations that this proposal does not address, as they were not considered in the stakeholder process and no use case demonstrating how such a facility may participate currently exists.

PJM Transmittal Letter, Docket No. ER25-1095 (Jan. 29, 2025).

4. Technical Resource Quality (15% weight): The maximum capacity factor achievable by renewable resources at the site location, ensuring that replacement generation would maintain high productivity levels.

5. Site Development Suitability (5% weight): The percentage of non-urbanized land area surrounding the plant, reflecting the practical feasibility of large-scale renewable infrastructure development.

6. Plant Interconnection Capacity (10% weight): The existing thermal plant's megawatt capacity, prioritizing larger facilities that offer more substantial grid interconnection infrastructure for renewable integration.

Ranking criteria for surplus interconnection development at renewables power plant sites:

1. Resource Quality Performance (30% weight): The capacity factor of the existing renewable facility, where higher performing sites demonstrate proven resource quality suitable for expansion.

2. Expansion Potential (30% weight): The absolute megawatt capacity of additional renewable resources available within a 6-mile radius, quantifying the scale of potential capacity additions.

3. Economic Optimization Opportunity (20% weight): The ratio between theoretically optimal capacity factor and current operational performance, identifying sites where technological upgrades or repowering could yield significant improvements.

4. Site Development Viability (10% weight): The percentage of non-urbanized land area surrounding the facility, assessing the feasibility of physical expansion without urban development constraints.

5. Existing Plant Scale (10% weight): The current installed capacity of the renewable facility, recognizing that larger existing plants have demonstrated interconnection value and operational viability at scale.

C. [additional section to potentially be added based on Abe/Roselle vetting these opportunities with developers]

VI. Pathways to implement surplus interconnection (Roselle)

Utilities can facilitate the connection of new energy resources using surplus interconnection through multiple pathways. They can contract with third-party or affiliated entities that use surplus interconnection through power purchase agreements, or purchase and sale agreements. Alternatively, utilities may themselves develop new surplus interconnection resources. This section describes best practices used in procurement of resources that use surplus interconnection service, and provides recommendations to utilities, regulators and legislators to implement these practices.

[roadmap summary of the various issues]

A. Utilities should make surplus interconnection service available to RFP bidders, and provide details on surplus interconnection opportunities within their system.

Requests for proposals vary significantly in the extent to which they explain or clarify the opportunity for bidders to use surplus interconnection service.

1. *Utilities can provide clear guidance on surplus interconnection opportunities within the context of an all-source RFP*

For example, in its 2024 All-Source RFP, OG&E explained that "OG&E Surplus Interconnection service is being made available by the Company for this RFP to expand opportunities for prospective Bidders."[29] OG&E identified three specific generation facilities that presented surplus interconnection opportunities, explaining that each facility had between 100 and 300 MW of surplus interconnection service available.[30] It provided for bidders to "obtain the OG&E Surplus Interconnection Information for the relevant OG&E Generation Facilities after signing a Mutual Nondisclosure Agreement."[31] Further, OG&E provided assurance to bidders that it "will cooperate with any successful Bidder . . . using OG&E Surplus Interconnection . . . to implement a Project and to obtain necessary interconnection studies and Permits."[32]

By contrast, some RFPs make it challenging for bidders to use surplus interconnection service, even where projects using surplus interconnection are technically eligible. For example, Pacificorp's 2022 All-Source RFP did not include any terms specifically discussing surplus interconnection.[33] While Pacificorp clarified in response to questions from potential bidders that surplus interconnection studies would be accepted as a form

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[29] OG&E 2024 All Source Request for Proposals, at 37, Appendix H.
[30] *Id.*
[31] *Id.*
[32] *Id.*
[33] *See* Pacificorp, 2022 All-Source Request for Proposals (Apr. 29, 2022), available at https://www.pacificorp.com/content/dam/pcorp/documents/en/pacificorp/suppliers/rfps/pacificorps-2022-all-source-request-for-proposals/PacifiCorp_2022AS_RFP_Main_Document.pdf.

of eligible documentation from bidders seeking to meet the RFP's interconnection-milestone requirements, it also explained that "PacifiCorp is not offering to potential third-party bidders for potential surplus interconnection service the interconnection facilities supporting its generating facilities."[34] [Thus, in practice, the opportunity to use surplus interconnection in serving the articulated need was restricted to Pacificorp's own affiliates.]

Note that projects may also use surplus interconnection at existing generating facilities that are not owned by the utility issuing the RFP. This will most often occur in RTO regions, where delivery to the utility does not depend on interconnecting to the utility's affiliated transmission system. But it may also occur in cases where facilities owned by third parties are connected to the utility's transmission system, as occurs in the case of power purchase agreements. In such cases, the utility should provide guidance that renders these projects eligible as well.[35] Further, because existing power purchase agreements often require amendment in order to accommodate surplus interconnection service, the utility can increase the pool of potentially available surplus interconnection by indicating that it will cooperate in renegotiating any relevant offtake agreements as necessary in order to accommodate surplus interconnection service.

2. *Utilities can issue RFPs targeting particular surplus interconnection opportunities*.

Another approach that has been used successfully is to issue RFPs targeting specific surplus interconnection opportunities. For example, Southwestern Electric Power Company issued an RFP in 2024 that specifically targets "new BESS facilities at the Harry D. Mattison power plant site," which offered "the opportunity to capture up to 210 MW of incremental capacity by using SPP's surplus interconnection service process."[36]

B. In issuing RFPs, utilities should offer clear guidelines for bidders that plan to use surplus interconnection service.

Topics that should be addressed include: (1) any required interconnection milestones; (2) eligible contract structures, . . .

1. *Interconnection-related requirements*

Different RFPs take different approaches to evaluating resources proposing to interconnect with surplus. Some RFPs require respondents to have completed the surplus interconnection study process,[37] whereas others allow for contracts to be awarded to

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[34] Pacificorp 2022 All Source RFP – Questions and Answers, at 2, 5, 10, available at

[35] For example, in OG&E's 2024 RFP, it explains that "[i]f the Project will use surplus interconnection service to which OG&E is not a party," the bidder should "indicate the capacity of the original and surplus requests and describe the availability of transmission capacity on a continuous or scheduled basis." OG&E 2024 All Source Request for Proposals, at 12.

[36] American Electric Power Service Corporation as agent for Southwestern Electric Power Company, Request for Proposals Purchase and Sale Agreements (PSAs) (Jan. 2024), at 9, available at https://www.swepco.com/lib/docs/business/b2b/rfp/swepco/SWEPCO_PSA_RFP_Final_1-31-24.pdf.

[37] *See, e.g.*, Pacificorp 2022 All-source RFP

projects prior to submitting a surplus interconnection request where such projects demonstrate a pathway to surplus interconnection.[38]

This second, more permissive approach opens up RFP opportunities to a broader range of potential respondents, increasing the range of potential competitors available to take full advantage of surplus interconnection service. While it entails some risk that the grid operator's analysis of a surplus interconnection request will identify unanticipated upgrades, that risk can be managed by requiring applicants to set forth "expected new interconnection facilities and network upgrades within the [proposed s]ite."[39] A more permissive approach may be warranted for surplus interconnection in part because projects using surplus service may often move from interconnection agreement to commercial operations more quickly.

2. *Contract structures*

[This section will explain how RFPs may be structured to require Purchase and Sale Agreements for projects using surplus, rather than Power Purchase Agreements, for cases where the utility owns the existing plant. Such a structure allows the utility owner of the existing resource to manage the operations of the new surplus interconnection resource together with the existing resource.]

[Note that generally speaking, the renewable power plants that present a surplus interconnection opportunity are owned by merchant developers, whereas thermal power plants that present surplus interconnection opportunities are owned by incumbent utilities]

C. Utilities can pursue surplus interconnection opportunities on their own, outside the context of RFPs.

[Self build proposals evaluated alongside RFPs[40]]

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[38] *See, e.g.,*, OG&E 2024 All Source Request for Proposals, at 37, Appendix H.
[39] OG&E 2024 All Source Request for Proposals, at 16 n. 7.
[40] *See* American Electric Power Service Corporation as agent for Southwestern Electric Power Company, Request for Proposals Purchase and Sale Agreements (PSAs) (Jan. 2024), at 1 ("The Company plans to evaluate one or more Self-Build Proposals in addition to the Proposals received in the PSA RFP. The Self-Build Proposals will be developed by the AEP Projects Group who are independent from the RFP Team. Self-Build Proposals shall 1) be submitted to the RFP Team one day prior to the PSA, PPA, and CPA Proposal Due Date, and 2) use as a basis for the Proposal, the information required in the PSA Proposal Content Section (RFP Section 6) and all applicable appendices."), available at https://www.swepco.com/lib/docs/business/b2b/rfp/swepco/SWEPCO_PSA_RFP_Final_1-31-24.pdf.

[self build independent of any RFPs[41]]

[at the end of this section include utility recommended action, IURC action, and legislative action to implement these steps]

VII. Streamlining siting and permitting for supply resources that use surplus interconnection (Jennifer)

A. Background

In Indiana, the primary authority for both the siting and permitting of renewable energy projects is reserved for local governments (i.e., counties/municipalities).[42] The Indiana Office of Energy Development coordinates and administers state energy policies and federal grant programs, which the Indiana Utility Regulatory Commission oversees electric utility regulation.[43] Wind and solar facilities, as "alternate energy production facilities," are exempt from the requirement that the Indiana Utility Regulatory Commission must issue a Certificate of Public Convenience and Necessity before construction begins.[44] Instead, solar energy systems generally require a specific permit, such as a conditional use permit or a special exception permit, from the local board of zoning appeals.

Indiana State Code authorizes counties to regulate land use development, including renewable energy projects, through standards and processes in zoning ordinances.[47] Stated another way, local government entities oversee renewable siting and development processes, which are guided by local land use regulations.[48] All solar projects, regardless of size, are subject to local land use controls and must acquire zoning or development permits from the local government in which the project is located.[49] Siting regulation and land use permitting for wind energy development is similarly handled by local governments through local zoning.[50] Zoning ordinances may further require plans and studies as part of

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[41] [Otter Tail Hoot Lake Solar and Solway Solar]
[42] Regulatory Assistance Project, Laws in Order: An Inventory of State Renewable Energy Siting Policies at 32, June 2024, available at: https://eta-publications.lbl.gov/sites/default/files/rap-enterline-valainis-laws-order-inventory-state-renewable-energy-siting-policies-2024-june.pdf ("RAP Siting Report").
[43] *Id.*
[44] *See* Ind. Code § 4-4.1-13, § 8-1-8.5-7, and § 7(2).
[47] Ind. Code § 36-7-4-202.
[48] Great Plains Institute, Indiana Renewable Energy Guide: A Guide for Local Governments at 2, December 2020, available at: https://eri.iu.edu/documents/in-renewable-energy-guide.pdf.
[49] *Id.*
[50] *Id.*

the renewable energy approval process, which commonly includes economic development agreements, maintenance plans, transportation plans, and site plans. Planning and zoning are voluntary, with 82 of 92 counties adopting both planning and zoning. On a voluntary basis, some counties opt into a "area plan commissions", which carry out planning at a more regional scale across several counties within the state.[51]

SB 411 (enacted in 2022) created a voluntary program allowing counties and municipalities to opt into a default set of siting standards.[53] Upon adoption such standards, they may become certified as "solar energy ready communities" and/or "wind energy ready communities."[54] Beyond providing for a more uniform siting process across these communities, the program also renders certified community eligible to receive technical assistance and incentive funds from the Office of Energy Development.[55] Such funds may be granted for a period of 10 years at $1/MW hour of electricity generated.[56]

Bills have been put forth in recent Indiana legislative sessions to provide for greater state control over siting and permitting.[61] Although both legislative efforts ultimately failed, debate around these measures demonstrated the challenges that wind and solar projects face in obtaining necessary local permits.[62]

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[51] Two types of "plan commissions" exist in the state: (1) advisory plan commissions and (2) area plan commissions. See: Ind. Code § 36-7-4-405 and cite to this report: https://planning-org-uploaded-media.s3.amazonaws.com/documents/FINAL-CitizenPlannersGuide-3.20.17-Ch.1-PlanCommissionBasics.pdf. Advisory plan commissions are legislated to plan at a county, city, or town scale. 49 counties have an advisory plan commission for unincorporated portions of the county. Area plan commissions exist as more "regional" entities that serve both cities and counties. 33 counites participate in area plan commissions, operating as cooperative efforts between the county and at least one municipality, including staff.

[53] See: SB 411 language here: https://legiscan.com/IN/text/SB0411/2022; and Ind. Code § 8-1-41-4 for default standards for "wind energy ready communities" and Ind. Code § 8-1-42-6 for default standards for "solar energy ready communities".

[54]

[55] Cite to: https://www.in.gov/oed/files/Energy-Ready-Communities-Handout.pdf and https://csgmidwest.org/wp-content/uploads/2024/01/2023_MLC_Renewable-Energy-Siting-Laws-Brief.pdf.

[56] *Id.*

[61] HB 1381 (2022) and HB 1628 (2025).

[62] *See, e.g.*, David Adams and George Lepeniotis, *Who Will Control the Future of Alternative Energy Projects in the State of Indiana* (Feb. 19, 2025), available at https://www.kriegdevault.com/insights/who-will-control-the-future-of-alternative-energy-projects-in-the-state-of-indiana (discussing HB 1628); London Gibson and Sarah Bowman, *After local pushback, Indiana bill on wind, solar standards take a '180 degree turn'*, available at https://www.indystar.com/story/news/environment/2021/04/02/indiana-energy-bill-wind-solar-power-standards-environment/4840386001/ (discussing HB 1381).

B. Recommendations

Surplus interconnection presents a particularly compelling case for a streamlined siting and permitting process because, by their nature, surplus interconnection projects are built on or near to the sites of existing power plants. As such, they are inherently less likely to pose land use impacts opposed by communities. Further, the points of interconnection for surplus interconnection projects are known in advance.

[Options for streamlining siting and permitting for surplus interconnection projects include:

- Vest siting and permitting authority for surplus interconnection projects with the IURC (i.e. mirror HB 1628, but with a different eligibility threshold. Rather than having the state process apply to all "multi-jurisdictional infrastructure projects," provide for it to apply to all projects that make use of surplus interconnection service.
    - Potentially in combination with this approach, commission an analysis of all potential surplus interconnection sites and put forth specific smart siting principles for these sites (e.g. create "development zones" within these sites, as well as areas to avoid). Provide that so long as projects develop according to these principles, their state permits would be automatically granted on a fast timeline (e.g. 30 days from the time the application is made).
- Provide for backstop siting authority for surplus interconnection projects, and provide for minimum standards that local siting standards must meet for all surplus projects, potentially mirroring the default siting standards that were established pursuant to SB 411.

- Establish permitting timeline for surplus interconnection projects that are reasonable and transparent.[63] Depending on whether enacted in conjunction with the ideas above, these timelines could be applied to either the local or state process and could require action by a set date (e.g. require denial or approval within 45 days of the permitting application being submitted to the agency).]

[63] According to a recent report from the Regulatory Assistance Project ("RAP"), 31 states have timelines governing the process of issuing project permits. RAP Siting Report at 10.

VIII. Economic development incentives for surplus interconnection development (Narayan)

Indiana's economic development framework is already built around energy as a competitive advantage. The Indiana Economic Development Corporation (IEDC), created in 2005, is charged with "coordinat[ing] state economic development resources to attract and retain jobs in Indiana" and "assisting communities in preparing sites for new investment."[75] The Office of Energy Development (OED), created in 2009, is similarly directed to "develop and implement comprehensive energy policies that support reliability, stability, affordability, and environmental sustainability of electric service" while coordinating with other state agencies.[76] OED spearheads efforts like the Energy-Ready Communities certification helping local governments incorporate solar and wind into their planning processes. Together, these agencies are laying the groundwork for energy-informed economic development, aligning site readiness and workforce strategies with the state's energy needs.

Indiana's major utilities also play a central role in economic development. Duke Energy's "Site Readiness Program" works with local governments to prepare parcels for large industrial projects, evaluating everything from transportation access to available grid infrastructure. Hoosier Energy maintains an economic development team that provides communities with site marketing support and touts the benefits of reliable, affordable power as a business advantage. AES Indiana and NIPSCO similarly coordinate with state and local leaders to identify priority sites and facilitate new business investment. These utility-led programs are natural vehicles for highlighting surplus interconnection capacity helping prospective companies site facilities where grid access is strongest and fastest to secure.

Despite these initiatives run by the state and utilities, there remain many opportunities to better leverage existing authorities and programs to incentivize surplus interconnection. Indiana already deploys site readiness grants, tax abatements, workforce training funds, and utility-led site marketing as part of its economic development strategy. Yet none of these tools explicitly consider where the electric grid has surplus capacity. Aligning economic development incentives with surplus interconnection opportunities would allow Indiana to accelerate clean energy deployment and avoid costly transmission upgrades, all while reinforcing the state's reputation for affordable and reliable power.

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[75] *See* Ind. Code § 5-28-2-2.
[76] *See* Ind. Code § 4-3-23-5.

The IEDC's statutory authorization is broad enough to allow the agency to incorporate surplus interconnection into its site selection and incentive programs. IEDC is directed to coordinate state economic development resources and prepare communities for new investment—responsibilities that naturally encompass prioritizing sites with available grid capacity.[77] Likewise, OED has authority to account for surplus interconnection in statewide energy planning and workforce initiatives.[78] The Indiana Utility Regulatory Commission (IURC), through its authority to ensure that utilities provide reasonably adequate service at just and reasonable rates, could also direct utilities to highlight surplus-capacity sites in their economic development programs and filings.[79] Together, these entities could integrate surplus interconnection into economic development practice without any statutory changes.

At the same time, modest statutory refinements, for example, expressly directing IEDC and OED to consider surplus interconnection in their mandates could make this priority clearer and more durable. Just as HEA 1007 embedded the "five pillars" into IURC oversight and SB 411 created "renewables ready" communities, the General Assembly could clarify IEDC's and OED's authority to integrate surplus interconnection into their mandates.

IX. Summary

[Include a summary table, either here or up front, which shows each specific recommendation (e.g. evaluate surplus opportunities, streamline siting and permitting, etc.) along with columns showing how that recommendation can be implemented (i) by the utility/existing gen owner; (ii)state commission; and (iii) legislature

[77] *See* Ind. Code § 5-28-1-1(b)(2)–(3) (General Assembly finds the IEDC should "possess broad powers designed to maximize the state's economic development efforts" and will "lead the state's economic development efforts."), and § 5-28-5-2 (granting IEDC "all powers necessary or appropriate to carry out the corporation's public and corporate purposes."); *see also* § 5-28-28.4 (establishment of "Shovel Ready Site Development Center to "enable political subdivisions to obtain. . . permits to create sites that are ready for economic development.").

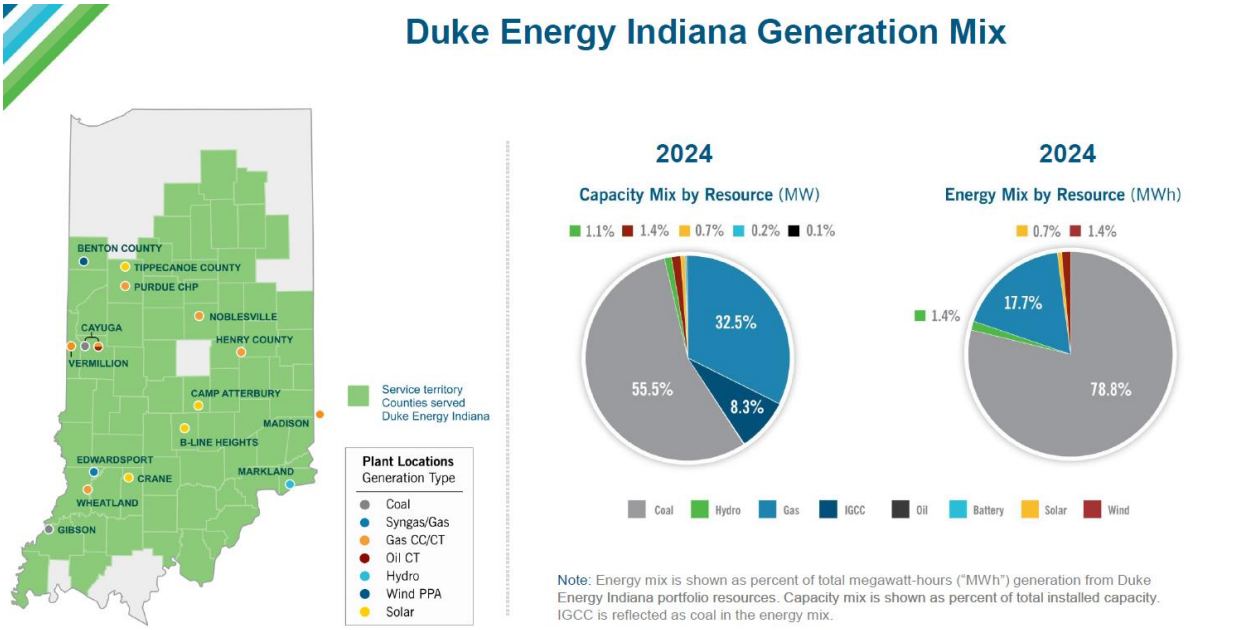
[78] *See* Ind. Code § 4-3-23-5 (directing OED to "develop and implement comprehensive energy policies that support reliability, stability, affordability, and environmental sustainability of electric service").

[79] *See* Ind. Code § 8-1-2-4 (requiring public utilities to furnish "reasonably adequate service and facilities" and prohibiting "unjust or unreasonable charge[s]").

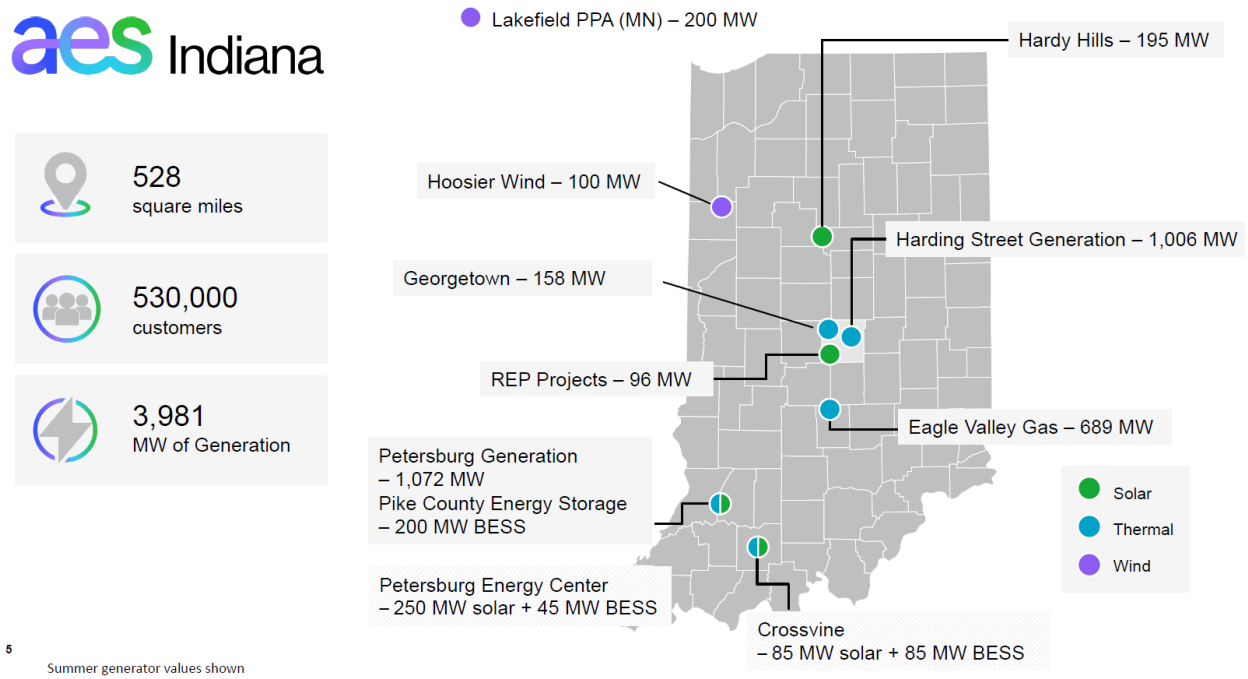
Appendix: Indiana Utility's Supply–Demand Balance

<u>Duke Energy Indiana (MISO)</u>:

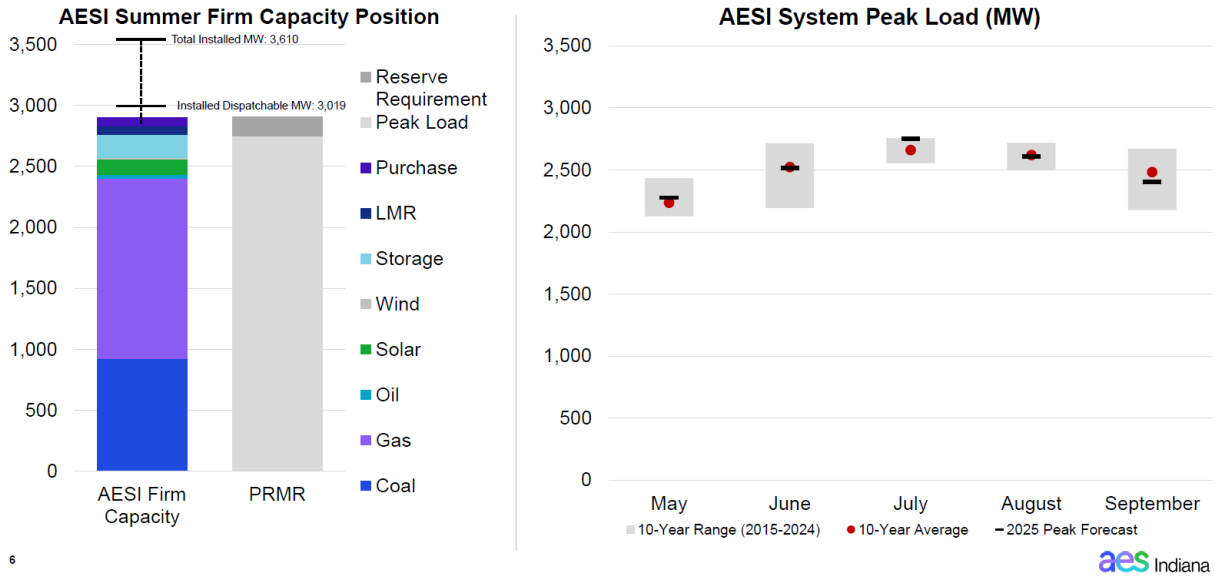
- DEI is largely flat to its summer peak load, purchasing 133 MW of 2.1 % of its total capacity for Summer 2025 from the market.
- Moving to DLOL improves DEI's accreditation values.



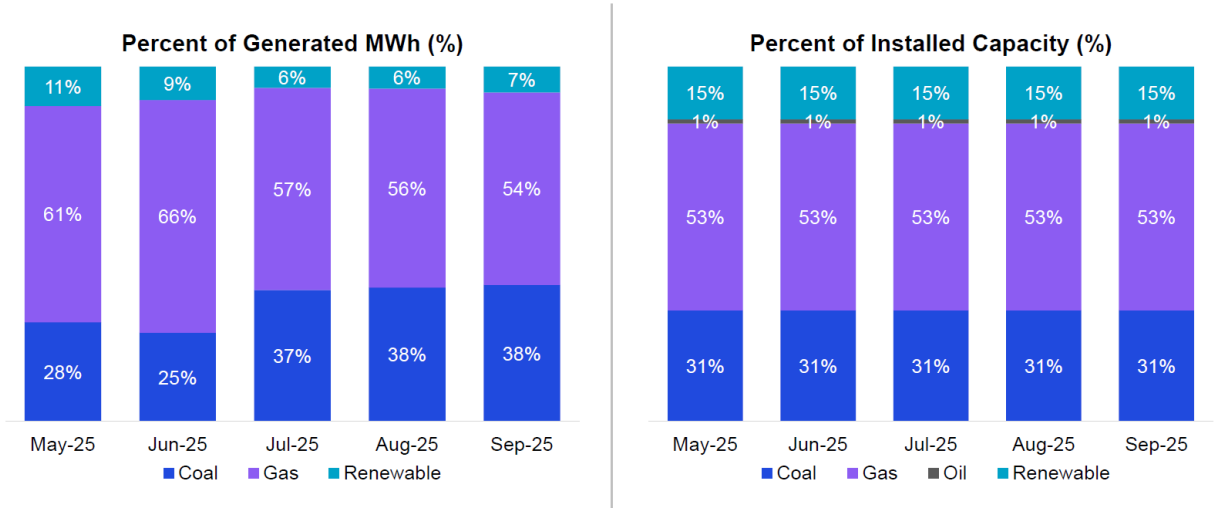
<u>AES Indiana (MISO):</u>



Summer load and supply summary

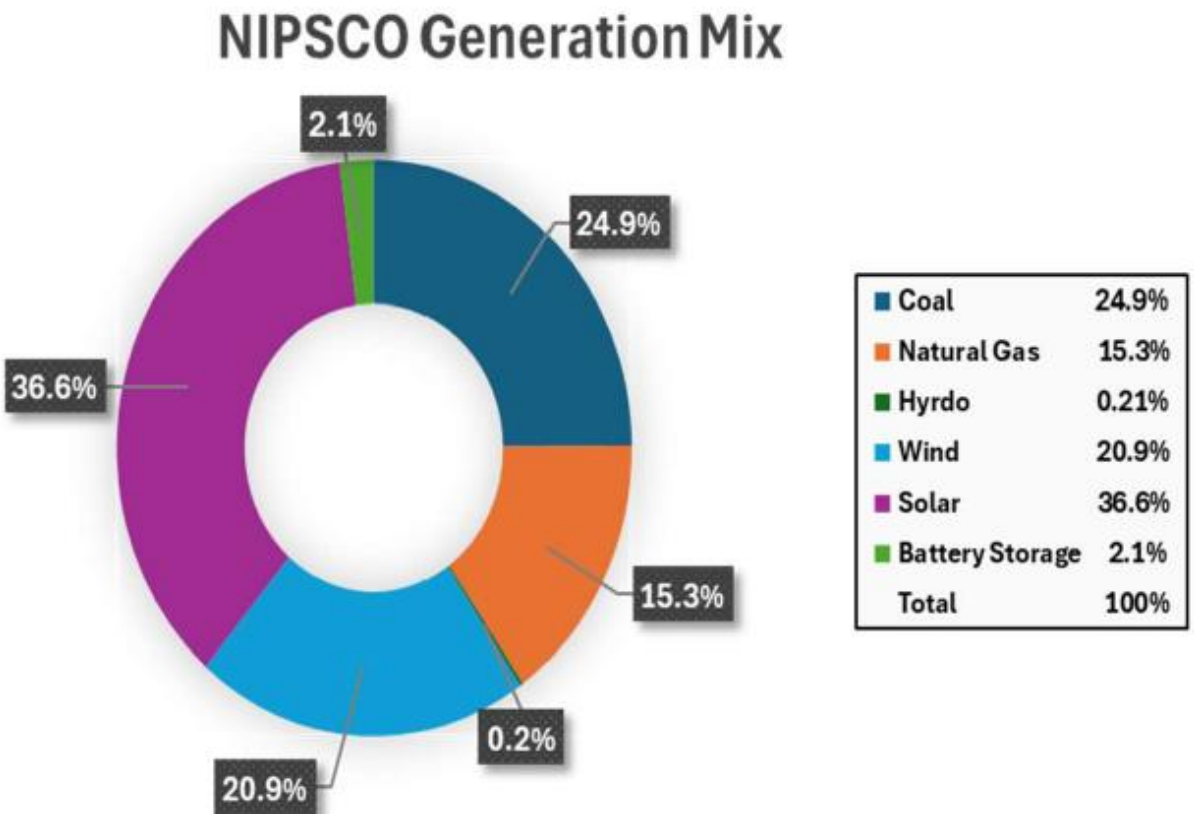


Diversified Fleet will Provide Reliability and Price Stabilization for Customers in Summer 2025



NIPSCO:

- NIPSCO participated in MISO's PRA with surplus capacity of 147 MW in the summer season and 252 MW in the fall season.



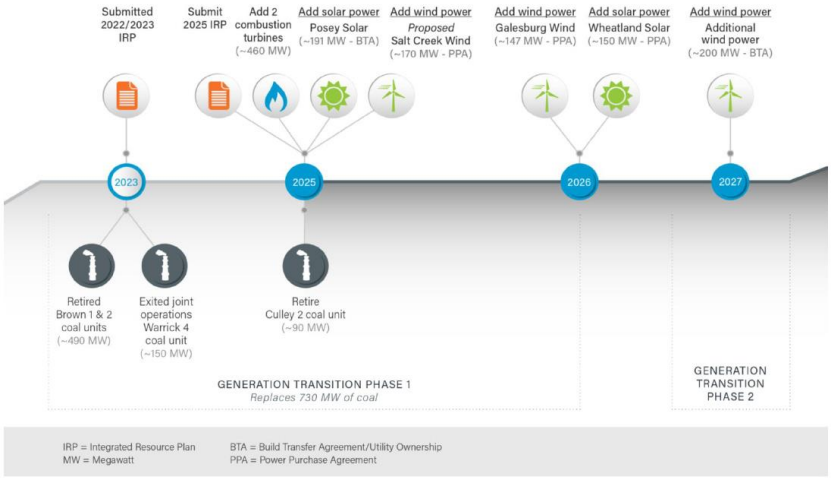
- 3,430 MW generating capacity
- – 2 coal generation facilities
- – 1 CCGT
- – 2 hydro electric generation facilities
- – 6 renewable generation facilities in service

SRF materials outline accredited capacity coverage for summer; its 2024 IRP updates portfolio timing amid rising large-load interest (including data centers). Recent filings also seek a **gas peaker CPCN** to firm the stack. (Indiana.gov, IURC Online Services Portal)

CenterPoint Indiana South (MISO) (CEI):

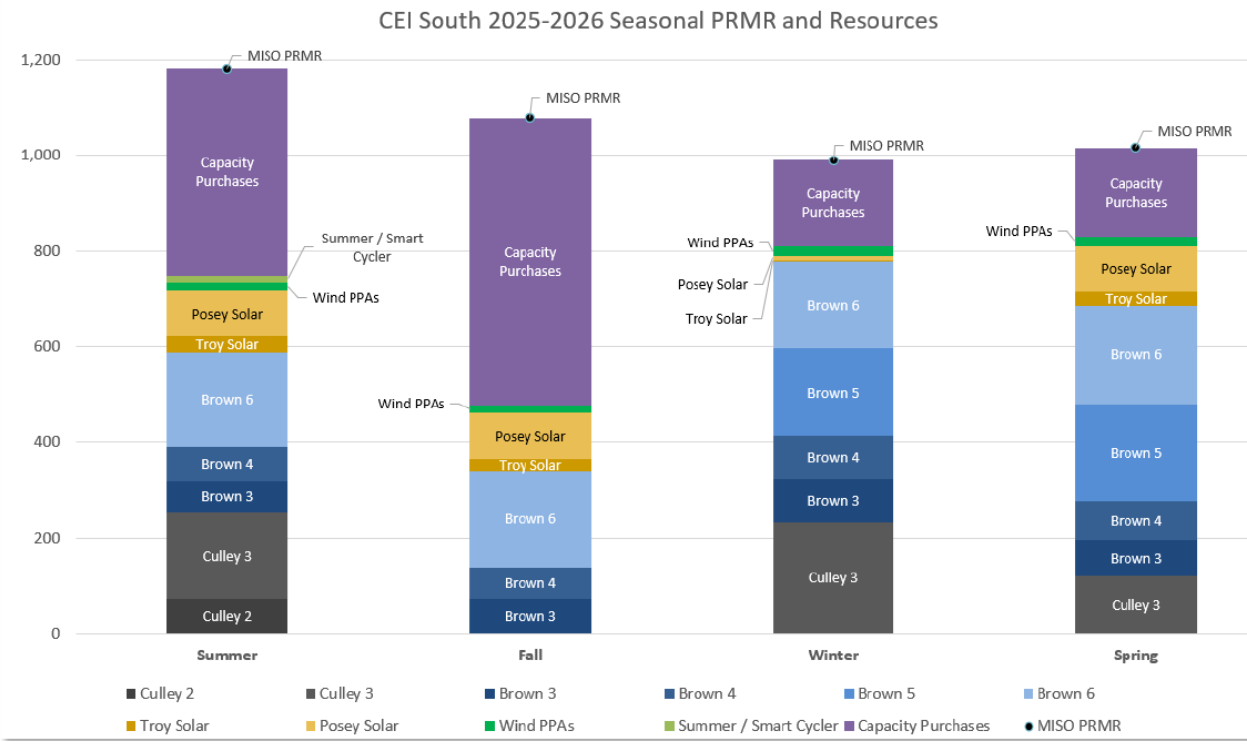
Indiana Electric Generation Transition: Roadmap

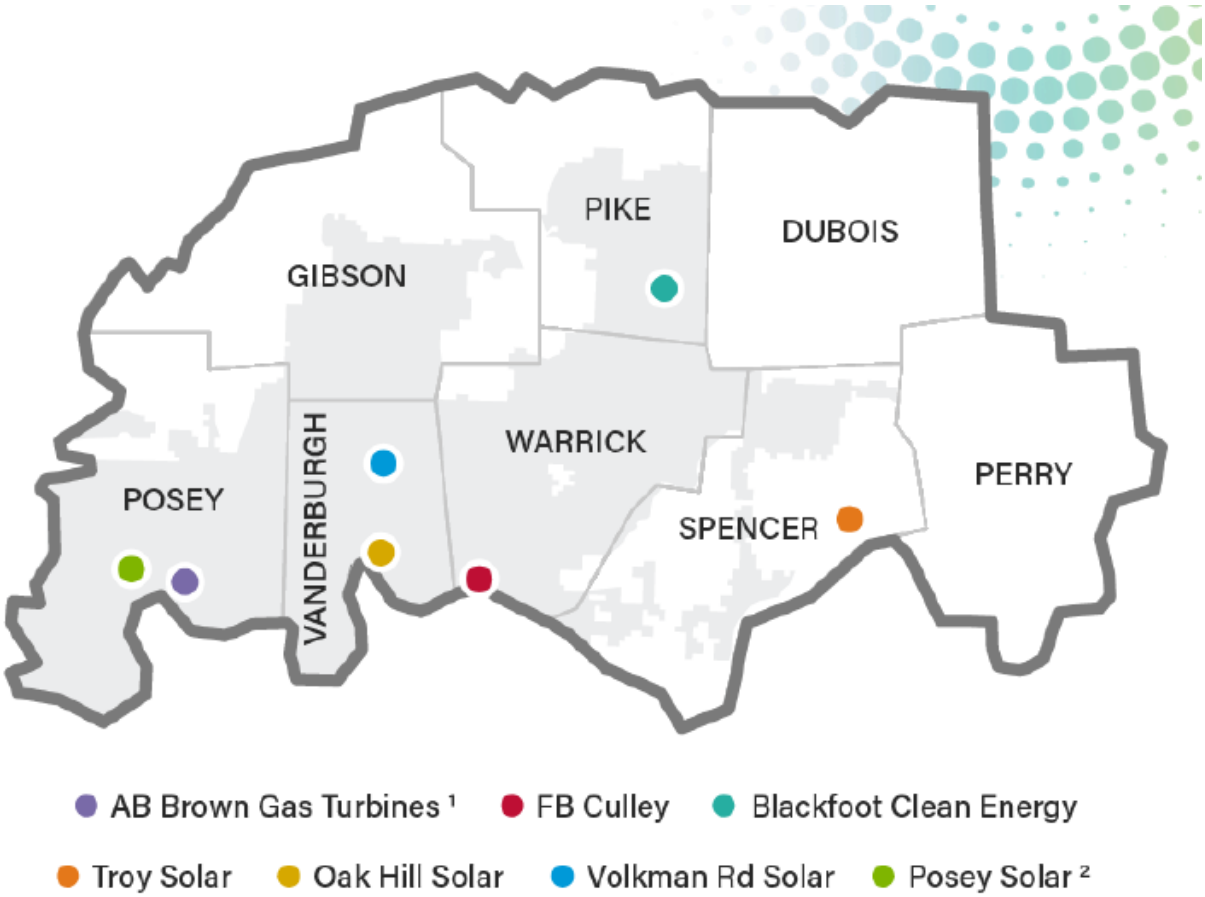
CEI South's generation resource diversification accounts for intermittent resource variability to meet customer needs



- Coal Units – 392 MW
  - ~ 29%FB Culley Units 2 and 3 – 360 MW
  - Ohio Valley Electric Cooperative (OVEC) – 32MW
- Gas Peaking Units – 620 MW
  - ~ 46%AB Brown Units 3 and 4 – 160 MW
  - AB Brown Units 5 and 6 – 460 MW
- Renewables – 329 MW
  - ~ 25%Posey Solar – 191 MW
  - Troy Solar – 50 MW
  - Volkman Road Solar – 2 MW (Includes 1 MW/4 MWHr Battery)
  - Oak Hill Solar – 2 MW
  - Blackfoot Landfill Gas – 3 MW
  - Fowler Ridge Wind PPA – 50 MW

- Benton County Wind PPA – 30 MW





Note: Fowler Ridge & Benton County Wind Farms not shown.
¹ABB CTs 5&6 expected to be in-service Q2 2025.
²Posey Solar expected to be in-service Q2 2025.

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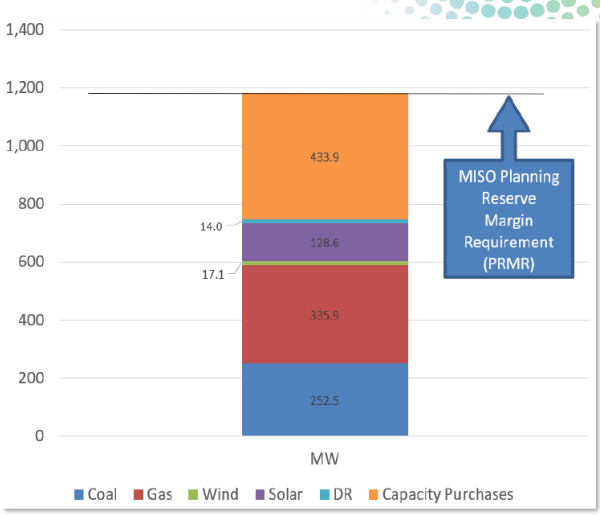
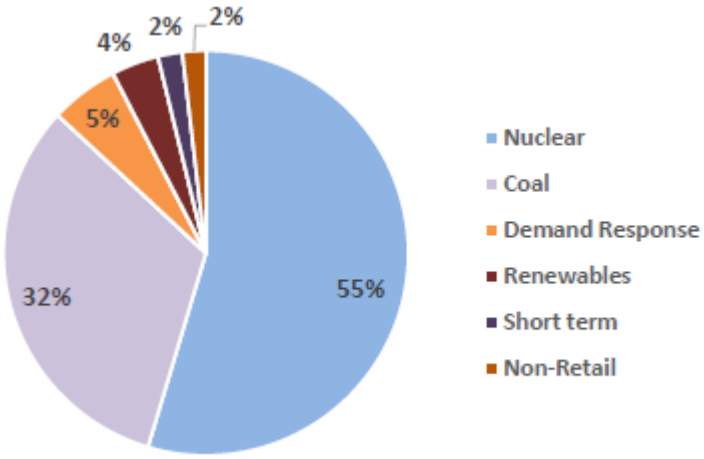
CEI South Capacity Resources for 2025 – 2026 Summer Season

- **CEI South Accredited Capacity:** 732 MW
  - Coal – 253 MW
  - Natural Gas – 336 MW
  - Solar – 129 MW
  - Demand Response *(Summer / Smart Cycler)* – 14 MW
- **Other Accredited Capacity:** 451 MW
  - Wind – 17 MW
  - OVEC – 30 MW
  - Capacity Purchases – 404 MW
- **Behind the Meter Capacity:** 5 MW
  - Blackfoot Renewable Gas – 3 MW
  - Volkman Road Solar (BTM) – 1 MW
  - Evansville Urban Solar (BTM) – 1 MW



Indiana Michigan Power (PJM):

- I&M's capacity obligation for summer 2025 is approximately 3,800MW
- Over 90% of I&M's capacity requirement for the summer of 2025 will be met by dispatchable generation resources
  - D.C. Cook Nuclear Plant
  - Rockport Unit 1
  - Coal and Natural Gas Contracts
- Demand Response accounts for approximately 200MW of capacity for the summer of 2025
- Existing wind purchase power agreements, owned solar facilities, and the run of the river hydro units account for approximately 145MW of accredited capacity

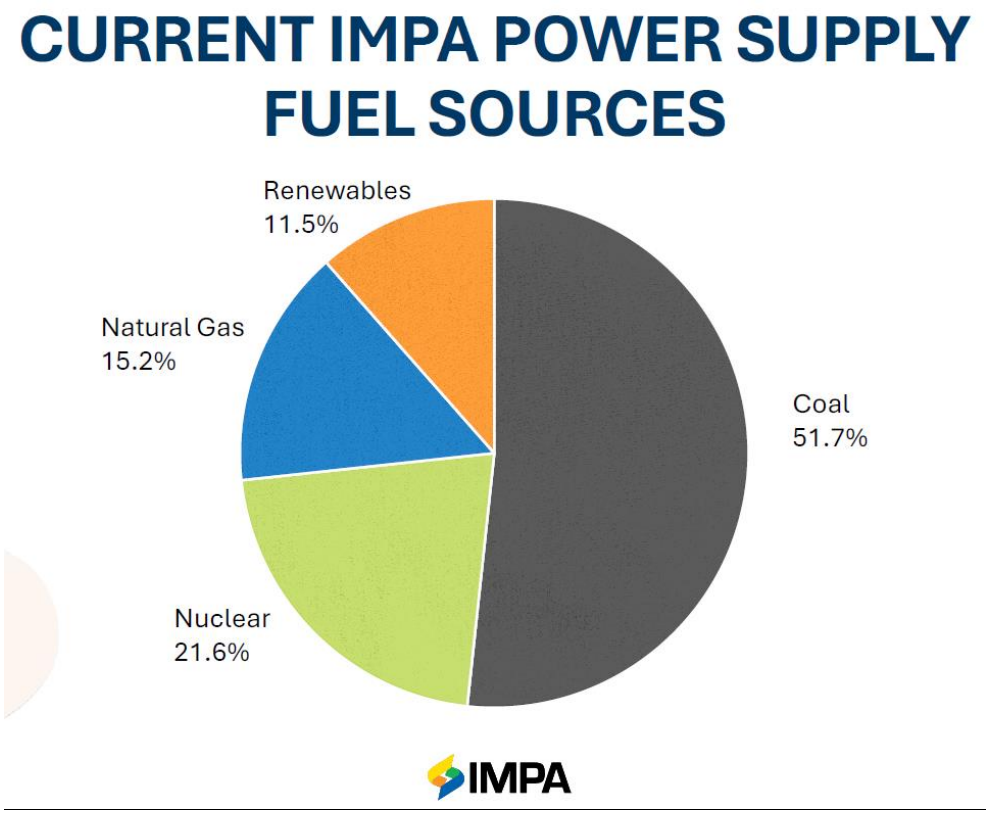


Resource Type	2025 UCAP (MW)
Nuclear	2,082
Coal	1,230
Demand Response	208
Renewables	145
Short term	75
Non-Retail	71
Total	**3,810**

- Filed a new **2024 Indiana IRP (Mar 28, 2025)**, planning to meet PJM reserve margin requirements as Rockport coal exits and Cook nuclear remains central. PJM's 2025/26 auction results (price spike vs. prior year) underscore a **tightening PJM margin**, shaping I&M capacity strategy. (Indiana Michigan Power, PJM)

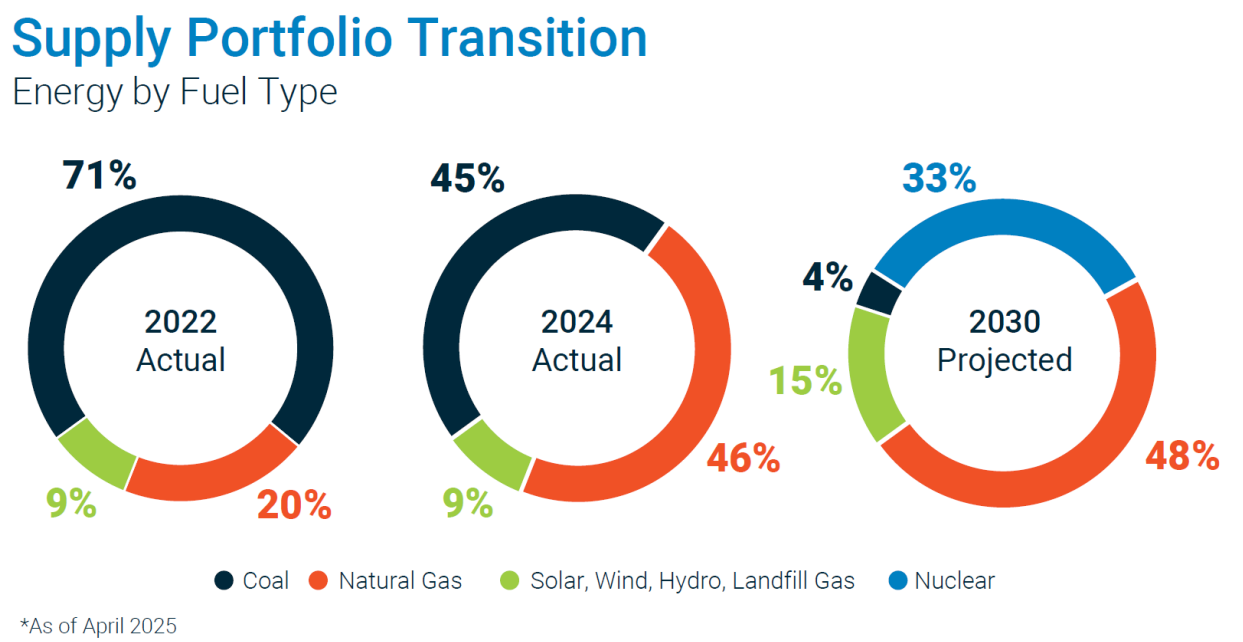
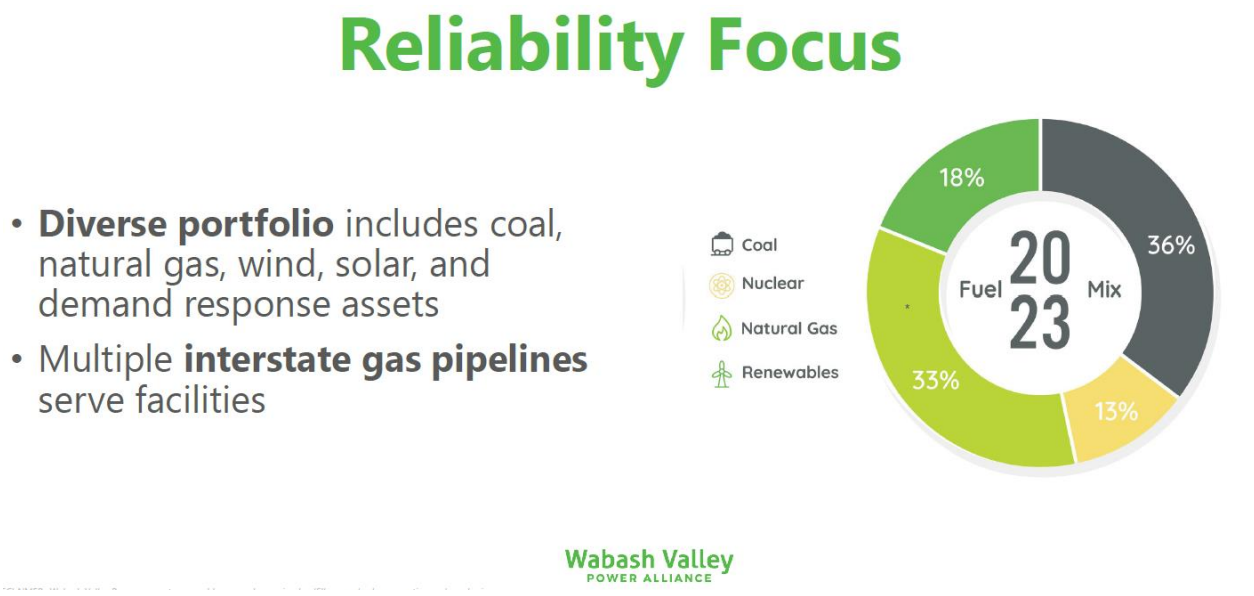
Indiana Municipal Power Authority (MISO/PJM):

- 1200 MW system load
- IMPA was formed as an Indiana joint action agency in 1980 by 11 communities & currently at 61 members



<u>Hoosier Electric (PJM/MISO:</u>

- G&T cooperative founded in 1949
- All-time system peak - 1,828 MW



Wabash Valley Power Alliance: